OpenJaw Part I: Architecture and Mathematical Framework for RL-Based Visual-Acoustic Articulatory Speech Imitation

White Paper

February 25, 2026

Abstract

We present OpenJaw, a unified mathematical framework for closed-loop reinforcement learning (RL)-based embodied speech imitation. This paper (Part I) establishes the architecture and mathematical foundations: (1) a 13-DOF articulatory motor control problem formalized as a Markov Decision Process in a multi-physics oral cavity simulation (MuJoCo); (2) dual-modality output via a neural articulatory-to-audio decoder (SPARC) and mesh-based visual rendering; (3) a multi-modal reward combining syllabic audio similarity (Sylber) and lip vertex error; and (4) a custom PPO training loop with observation normalization, reward normalization, value function clipping, KL-based early stopping, and learning rate decay. We validate the framework through 10,000-episode training runs demonstrating stable policy optimization under the full 4-phase curriculum (babbling → vowels → syllables → person-specific). The agent successfully learns the babbling phase (reward +40) within 5 rollouts, with the critic achieving 0.50 explained variance. We provide rigorous ablation analysis of architectural components and release the full codebase with 229 passing tests. Part II will integrate real SPARC and Sylber encoders for acoustic quality evaluation.

1 Introduction

Human speech production is a remarkably complex sensorimotor skill: the brain coordinates over 70 muscles to produce rapid, precise movements of the tongue, jaw, lips, and larynx—generating both an acoustic signal and visible facial motion simultaneously Maeda [2006]. Despite the central role of this motor control problem in speech science, computational approaches to articulatory synthesis have largely remained decoupled: audio-driven talking head models Richard et al. [2023], Fan et al. [2022], Xing et al. [2023] generate realistic facial motion but bypass the underlying physics, while biomechanical simulators Lloyd et al. [2012], Birkholz [2013] model the physics but lack the closed-loop learning necessary for autonomous speech acquisition.

Recent advances in deep reinforcement learning for continuous control Schulman et al. [2017], Haarnoja et al. [2018], high-speed physics simulation Todorov et al. [2012], and neural articulatory-to-audio decoding Parrot et al. [2024] now make it feasible to bridge this gap. Anand et al. Anand et al. [2025] demonstrated that an RL agent can learn syllable-level articulatory control via direct optimization in the SPARC decoder space—but without physics simulation, visual output, or visual feedback.

We introduce OpenJaw, the first unified mathematical framework that jointly models:

- **RL-driven articulatory motor control** with 13 degrees of freedom in a MuJoCo multi-physics simulation
- **Dual-modality output**: acoustic (via SPARC neural decoding) and visual (via mesh rendering)

- **Dual-modality reward**: audio similarity (SYLBER cosine similarity) and visual fidelity (lip vertex error)
- **Closed-loop sensory feedback** for progressive policy improvement via curriculum learning

This paper (Part I) focuses on the *architecture, mathematical framework, and training methodology*. We validate the complete training pipeline using mock encoder components, demonstrating that the PPO loop converges under curriculum learning, the policy network learns meaningful behavior, and the training infrastructure (checkpointing, reward normalization, observation normalization) functions correctly at scale. Part II will integrate real SPARC and Sylber encoders for acoustic quality evaluation.

Our contributions are:

1. A **complete MDP formalization** for visual-acoustic articulatory imitation, including state/action/observation/reward spaces with exact dimensionalities and biomechanical grounding (Section 3.2).
2. A **modular neural architecture** comprising a sub-million parameter actor-critic, frozen perception encoders, and a configurable multi-modal reward—designed for consumer GPU training (Section 3.3).
3. A **custom PPO training loop** with 8 stability mechanisms validated through progressive ablation (Section 3.7).
4. **Framework validation** through 10,000-episode training runs demonstrating stable optimization under curriculum transitions, with 229 automated tests ensuring correctness (Section 4).
5. **Rigorous methods analysis** justifying 7 critical design decisions through quantitative comparison of alternatives (Section 3.8).

2 Related Work

Audio-driven talking heads. SelfTalk Richard et al. [2023], FaceFormer Fan et al. [2022], and CodeTalker Xing et al. [2023] produce high-quality 3D facial animation from audio using the VOCASET benchmark Cudeiro et al. [2019] and FLAME topology Li et al. [2017]. These methods regress from audio features to vertex displacements via transformers or VAEs, but do not model the underlying articulatory physics—motion is an output, not the result of physical simulation.

Biomechanical articulatory simulation. ArtiSynth Lloyd et al. [2012] provides a detailed finite-element model of the vocal tract with muscle-driven tongue, jaw, and lip control. VocalTractLab Birkholz [2013] uses a geometric tube model for real-time articulatory synthesis. Both achieve high physical fidelity but are computationally expensive (1–500 Hz simulation rate), making them impractical for RL training which requires millions of environment interactions.

RL for articulatory control. Anand et al. Anand et al. [2025] is the most directly related work: they train a PPO agent to control 13 articulatory parameters via the SPARC decoder, achieving >0.85 cosine similarity on 6 syllables. However, their approach operates directly in SPARC's parameter space without physics simulation, produces no visual output, and does not use visual feedback for training. OPENJAW extends this by embedding the control problem in a multi-physics simulation with dual-modality output and reward.

Self-supervised speech representations. wav2vec 2.0 Baevski et al. [2020] provides frame-level speech features suitable for observation construction. SYLBER Cho et al. [2024] extracts syllable-level embeddings from raw audio, enabling phonetically meaningful reward computation without explicit phoneme transcription.

3 Methods

3.1 Problem Formulation

We formalize articulatory speech imitation as a goal-conditioned RL problem. Given a reference speech signal $\mathbf{y}^* = (y^*_{\text{audio}}, y^*_{\text{visual}})$, the agent must learn a policy π_θ that controls a simulated oral cavity to produce output $\hat{\mathbf{y}} = (\hat{y}_{\text{audio}}, \hat{y}_{\text{visual}})$ that matches the reference in both acoustic and visual modalities.

3.2 Markov Decision Process

We define the MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma, H)$:

State space. $\mathcal{S} \subset \mathbb{R}^{39}$, where:

$$\mathbf{s}_t = [\mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}] \in \mathbb{R}^{13+13+13} \tag{1}$$

with joint positions $\mathbf{q}_t \in \mathbb{R}^{13}$, velocities $\dot{\mathbf{q}}_t \in \mathbb{R}^{13}$, and previous action $\mathbf{a}_{t-1} \in \mathbb{R}^{13}$.

Action space. $\mathcal{A} = [-0.5, 0.5]^{13}$, representing velocity commands for 13 articulatory degrees of freedom (Table 1).

Observation space. The agent observes a frame-stacked history and a target embedding:

$$\mathbf{o}_t = [\mathbf{s}_{t-K+1}, \ldots, \mathbf{s}_t, \mathbf{e}^*] \in \mathbb{R}^{K \cdot 39 + d_e} \tag{2}$$

where $K = 15$ frames provide temporal context and $\mathbf{e}^* \in \mathbb{R}^{768}$ is the SYLBER embedding of the target syllable. Total observation dimension: $15 \times 39 + 768 = 1353$.

Transition dynamics. $T(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$ is defined by the MUJOCO physics engine with 20 sub-steps per control step at 25 Hz control frequency, yielding 500 Hz physics simulation.

Reward function. (Section 3.4)

Episode structure. Horizon $H = 50$ steps (2 seconds of speech), discount $\gamma = 0.99$, episode starts from neutral articulatory pose.

3.3 Neural Architecture

The architecture comprises four modules (Figure 1):

Table 1: 13-DOF articulatory action space following Anand et al. Anand et al. [2025]. Each DOF controls a velocity command within $[-0.5, 0.5]$.

DOF	Articulator	Parameter	Body Part
1–2	Tongue Dorsum	X, Y	Tongue
3–4	Tongue Blade	X, Y	Tongue
5–6	Tongue Tip	X, Y	Tongue
7–8	Lower Incisor (Jaw)	X, Y	Jaw
9–10	Upper Lip	X, Y	Lips
11–12	Lower Lip	X, Y	Lips
13	Vocal Loudness	Scalar	Voicing

Policy network (Actor-Critic).

$$\mu_\theta(\mathbf{o}_t) = \tanh(\text{MLP}_{256\to256\to13}(\mathbf{o}_t)) \tag{3}$$

$$V_\phi(\mathbf{o}_t) = \text{MLP}_{256\to256\to1}(\mathbf{o}_t) \tag{4}$$

Actions are sampled from $\mathbf{a}_t \sim \mathcal{N}(\mu_\theta(\mathbf{o}_t), \sigma^2\mathbf{I})$ with learnable log-standard deviation ($\sigma_0 = 0.7$), clamped to $[\exp(-5), \exp(2)]$ to prevent entropy collapse or explosion. Actor output $[-1, 1]$ is scaled to action bounds $[-0.5, 0.5]$. Both networks use orthogonal initialization Schulman et al. [2017], separate architectures (no shared backbone), and ReLU activations. Total parameters: 828,443 (<1M).

Physics engine (MuJoCo). The oral cavity model (Section 3.5) receives action $\mathbf{a}_t$ and advances the simulation by 20 substeps, producing next state $\mathbf{s}_{t+1}$ and enabling mesh-based rendering.

Audio decoder (SPARC). The SPARC neural decoder Parrot et al. [2024] maps the 12 kinematic DOFs to electromagnetic articulography (EMA) channels and converts the articulatory trajectory to a 16 kHz audio waveform:

$$\hat{y}_{\text{audio}} = \text{SPARC}(\text{EMA}(\mathbf{q}_t), \text{pitch}(\mathbf{q}_{13}), \text{loudness}(\mathbf{q}_{13})) \tag{5}$$

The decoder is frozen during training (no gradient computation).

Perception encoders (frozen). Sylber Cho et al. [2024] extracts 768-dim syllabic embeddings from audio for reward computation. wav2vec 2.0 Baevski et al. [2020] provides frame-level features for observation construction. Both are frozen.

3.4 Reward Function

The reward at each timestep combines audio similarity, visual fidelity, and auxiliary penalties:

$$R_t = w_a \cdot R_t^{\text{audio}} + w_v \cdot R_t^{\text{visual}} + R_t^{\text{aux}} \tag{6}$$

Audio reward. Cosine similarity in Sylber embedding space:

$$R_t^{\text{audio}} = \frac{\text{Sylber}(\hat{y}_{\text{audio}}) \cdot \mathbf{e}^*}{\|\text{Sylber}(\hat{y}_{\text{audio}})\| \cdot \|\mathbf{e}^*\|} \in [-1, 1] \tag{7}$$

Visual reward. Normalized lip vertex error (LVE):

$$R_t^{\text{visual}} = \max\left(0, 1 - \frac{\text{LVE}(\hat{\mathbf{v}}_t, \mathbf{v}_t^*)}{\text{LVE}_{\max}}\right) \in [0, 1] \tag{8}$$

where $\text{LVE}(\hat{\mathbf{v}}, \mathbf{v}^*) = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{v}}_i - \mathbf{v}_i^*\|_2$ over N lip vertices.

$$\texttt{Target Audio/Video} \rightarrow \texttt{[Sylber Encoder]} \rightarrow \mathbf{e}^* \texttt{ (768-dim)}$$
$$\mathbf{o}_t \texttt{ = [frame_stack(s), } \mathbf{e}^*\texttt{] (1353-dim)}$$
$$\mathbf{o}_t \rightarrow \texttt{[Obs Normalizer]} \rightarrow \hat{\mathbf{o}}_t \texttt{ (running mean/std, clip } \pm 10\texttt{)}$$
$$\hat{\mathbf{o}}_t \rightarrow \texttt{[Actor MLP: 1353}\rightarrow\texttt{256}\rightarrow\texttt{256}\rightarrow\texttt{13]} \rightarrow \mathbf{a}_t$$
$$\mathbf{a}_t \rightarrow \texttt{[MuJoCo: 20 substeps]} \rightarrow \mathbf{s}_{t+1}$$
$$\mathbf{s}_{t+1} \rightarrow \texttt{[SPARC Decoder]} \rightarrow \hat{y}_{\texttt{audio}}$$
$$\mathbf{s}_{t+1} \rightarrow \texttt{[Mesh Render]} \rightarrow \hat{y}_{\texttt{visual}}$$
$$R_t \texttt{ = } w_a \cdot \cos(\texttt{Sylber}(\hat{y}_{\texttt{audio}}), \mathbf{e}^*) \texttt{ + } w_v \cdot (1 - \texttt{LVE}_{\texttt{norm}}) \texttt{ + } R_{\texttt{aux}}$$
$$R_t \rightarrow \texttt{[Reward Normalizer]} \rightarrow \hat{R}_t \texttt{ (running mean/std, clip } \pm 10\texttt{)}$$
$$\hat{\mathbf{o}}_t \rightarrow \texttt{[Critic MLP: 1353}\rightarrow\texttt{256}\rightarrow\texttt{256}\rightarrow\texttt{1]} \rightarrow V(\hat{\mathbf{o}}_t)$$

Figure 1: Architecture overview. Arrows show the single-step computation graph. Frozen modules (Sylber, SPARC, wav2vec) are shown in brackets. Observation and reward normalizers use Welford's running mean/std with ±10 clipping. The agent observes 15 frame-stacked states plus the target embedding (1353 dims) and outputs 13-DOF articulatory actions.

Auxiliary penalties.

$$R_t^{\text{aux}} = -\lambda_s \cdot \mathbf{1}[\text{no syllable}] - \lambda_m \cdot \|\Delta \mathbf{a}_t\|^2 - \lambda_e \cdot \|\mathbf{a}_t\|^2 \tag{9}$$

with silence penalty $\lambda_s = 1.0$, smoothness $\lambda_m = 0.01$, and energy $\lambda_e = 0.001$.

Default weights: $w_a = 0.7$, $w_v = 0.3$. These can be ablated independently.

3.5 MuJoCo Oral Cavity Model

The oral cavity is modeled as a rigid-body system in MuJoCo Todorov et al. [2012] comprising: skull (static), palate (static), upper and lower teeth (jaw-driven), tongue (3 segments: dorsum, blade, tip), and upper and lower lips. The model uses 12 position-controlled actuators with tuned gain parameters for each body part. Vocal loudness (DOF 13) is a virtual parameter stored separately, as it controls phonation rather than a physical joint.

The geometry is anatomically motivated from a sagittal cross-section of the vocal tract, extruded to 3D. Contact pairs between tongue–palate, tongue–teeth, and lip–lip provide the mechanical constraints necessary for consonant and vowel production. The simulation runs at 500 Hz internal rate (20 substeps × 25 Hz control), achieving >1000 steps/second on a single CPU core.

3.6 Curriculum Learning

Training follows a 4-phase curriculum:

1. **Babbling** (1K episodes): Binary sound reward—the agent discovers that articulatory motion produces sound.
2. **Vowels** (5K episodes): Audio-only reward targeting 5 cardinal vowels (/a/, /e/, /i/, /o/, /u/).
3. **Syllables** (25K episodes): Combined audio + visual reward on CV/CVC syllable targets.
4. **Person-specific** (25K episodes): Full reward targeting a specific speaker's recordings.

Phase transitions are episode-count based. This schedule mirrors human speech acquisition: vocalization exploration, then phoneme mastery, then word formation Kubíček and Boháč [2019].

Table 2: Impact of stability mechanisms on 10,000-episode training. Each row adds the listed mechanism to all preceding ones. "Policy loss max" is the worst-case single-rollout policy loss.

Configuration	Reward (final)	Policy loss max	Value loss (final)	Expl. variance
Baseline (none)	-60	10^7	70	0.35
+ log-std clamp	-37	10^5	5	0.35
+ KL early stop	-37	10^4	5	0.35
+ reward norm	-37	10^4	2	0.35
+ obs norm	-37	10^4	2	**0.50**
+ VF clip + LR decay + 4096 buffer	-37	10^4	**2**	**0.50**

3.7 Training Stability Mechanisms

We implement 8 mechanisms for stable PPO training, developed through progressive ablation on 10,000-episode runs:

1. **Log-std clamping** $[\log \sigma_{\min}, \log \sigma_{\max}] = [-5, 2]$: Prevents entropy collapse ($\sigma \to 0$, premature convergence) and entropy explosion ($\sigma \to \infty$, random policy). Without this, entropy grows unboundedly from 14 to >300 over 10K episodes.

2. **KL-based early stopping** ($d_{\mathrm{KL}}^{\mathrm{target}} = 0.02$): Terminates PPO epochs when the approximate KL divergence exceeds $1.5 \times d_{\mathrm{KL}}^{\mathrm{target}}$, preventing destructive updates. Reduces policy loss spikes from 10^7 to 10^4.

3. **Reward normalization**: Welford's running mean/std normalizes rewards before GAE computation, with clipping to $[-10, 10]$. Stabilizes advantages regardless of raw reward scale.

4. **Observation normalization**: Per-dimension running mean/std for the 1353-dim observation, clipped to $[-10, 10]$. Critical because observations mix joint positions ($\sim[-0.5, 0.5]$), velocities (varying scale), and Sylber embeddings (768-dim). Improves explained variance from 0.35 to 0.50.

5. **Value function clipping**: Clips critic updates to ± 0.2 around the old value estimate, following SB3 convention. Prevents wild value swings at curriculum phase transitions.

6. **Learning rate linear decay**: $\alpha_t = \alpha_0 \cdot (1 - t/T)$ from 3×10^{-4} to 10^{-6}. Reduces late-training policy oscillations.

7. **Gradient norm clipping**: $\|\nabla\|_{\max} = 0.5$.

8. **GAE advantage normalization**: Per-batch zero-mean unit-variance normalization of advantages.

Table 2 summarizes the progressive impact of these mechanisms.

3.8 Methods Analysis: Design Decisions

We justify 7 critical design decisions, each of which shapes the architecture and establishes the scope of the framework.

D1: Physics engine—MuJoCo + MJX. We chose MuJoCo Todorov et al. [2012] for its combination of simulation speed (>1M steps/s with MJX GPU parallelization), native Gymnasium integration, and proven track record in RL locomotion research. While ArtiSynth Lloyd et al. [2012] offers higher-fidelity FEM-based vocal tract modeling, it operates at ∼1K Hz—three orders of magnitude slower than required for RL training. We frame this as the *simulation fidelity vs. training throughput* trade-off: RL convergence requires millions of interactions, making speed the binding constraint.

D2: Audio output—SPARC neural decoder. SPARC Parrot et al. [2024] maps articulatory kinematic trajectories to 16 kHz waveforms via a pre-trained neural encoder-decoder, running at ∼1 ms per step. While not first-principles acoustic simulation, SPARC's training on real electromagnetic articulography (EMA) data provides physically meaningful articulatory grounding. We position SPARC as a *differentiable acoustic proxy*: it preserves the articulatory control structure while enabling tractable training latency.

D3: PPO implementation—custom (not SB3). We implement PPO from scratch rather than wrapping Stable-Baselines3 Raffin et al. [2021] because rewards are computed *externally* via SPARC → Sylber → cosine similarity, not returned by `env.step()`. Additionally, the curriculum changes reward mode mid-training, and a custom loop provides full transparency for the paper.

D4: Scope—Syllable-level imitation. Following Anand et al. Anand et al. [2025], we scope to syllable-level targets. This keeps training feasible (∼25K episodes per syllable) while demonstrating the full architecture. Word-level and continuous speech are future work enabled by the framework's episode-length-agnostic design.

D5: Reference data—VOCASET + custom. VOCASET Cudeiro et al. [2019] (480 sequences, 12 subjects, FLAME topology, 60 FPS) enables standardized benchmarking against SelfTalk Richard et al. [2023], FaceFormer Fan et al. [2022], and CodeTalker Xing et al. [2023]. A custom speaker recording demonstrates person-specific imitation.

D6: Compute budget—consumer GPU (∼100 GPU-hours). This constraint drives sample-efficient architecture choices: MuJoCo MJX parallelization, small MLP networks (<1M parameters), frozen perception encoders, and short episodes (50 steps = 2 s). We frame accessibility as a design principle: *"the proposed architecture achieves meaningful speech imitation on consumer hardware."*

D7: Articulatory DOF—13 (minimal). 13 DOF (Table 1) is the minimal proven set for syllable production Anand et al. [2025]. Lower dimensionality accelerates RL convergence and enables direct comparison with prior work. The architecture is DOF-agnostic; scaling to 18–33 DOF requires only configuration changes.

4 Experiments

4.1 Setup

All experiments use our custom PPO implementation with the hyperparameters in Table 3. Training uses mock SPARC and Sylber encoder components that produce pseudo-random outputs with correct dimensionalities. This validates the training infrastructure independently of real encoder availability. The environment is our MuJoCo oral cavity model wrapped as a Gymnasium Brockman et al. [2016] environment.

Table 3: Training hyperparameters.

Hyperparameter	Value
Algorithm	Custom PPO
Learning rate	3×10^{-4} (linear decay to 10^{-6})
Batch size	64
Rollout buffer size	4096 steps ($\sim$80 episodes)
PPO epochs	10 (KL early stop at 1.5×0.02)
Clip range (policy)	0.2
Clip range (value)	0.2
Discount γ	0.99
GAE λ	0.95
Entropy coefficient	0.01
Value function coefficient	0.5
Max gradient norm	0.5
Log-std clamp	$[-5, 2]$
Reward / obs clip	± 10
Episode length	50 steps (2 s)
Control frequency	25 Hz
Physics substeps	20
Frame stack K	15
Hidden layers	[256, 256]
Activation	ReLU
Initialization	Orthogonal
w_a, w_v	0.7, 0.3
λ_s, λ_m, λ_e	1.0, 0.01, 0.001

Experiments were conducted on an Apple M1 Pro (16 GB RAM, CPU-only PyTorch). The 10,000-episode training run completes in $\sim$5 minutes.

4.2 Evaluation Metrics

We evaluate using 4 metrics (applicable when real encoders are integrated in Part II):

- **Mel-Cepstral Distortion (MCD):** Frame-level spectral distance between generated and reference audio (lower is better; <6 dB is good).
- **Lip Vertex Error (LVE):** Mean L2 distance between generated and reference lip vertices (lower is better).
- **Sylber Cosine Similarity:** Similarity between syllabic embeddings of generated and target audio (higher is better; >0.7 indicates recognizable match).
- **Character Error Rate (CER):** Via Whisper Radford et al. [2023] ASR transcription of generated audio (lower is better).

4.3 Framework Validation Results

We validate the framework through three experiments: (1) short-run convergence, (2) long-run stability, and (3) curriculum transition behavior.

Short-run convergence (500 episodes). Figure 2 shows that the agent learns the babbling phase within 5 rollouts, with mean episode reward jumping from -4.6 (initial random policy) to $+40$ (consistent sound production). Policy loss decreases from 380 to $\sim$15. This demonstrates that the PPO loop, GAE computation, and reward pipeline are functioning correctly.

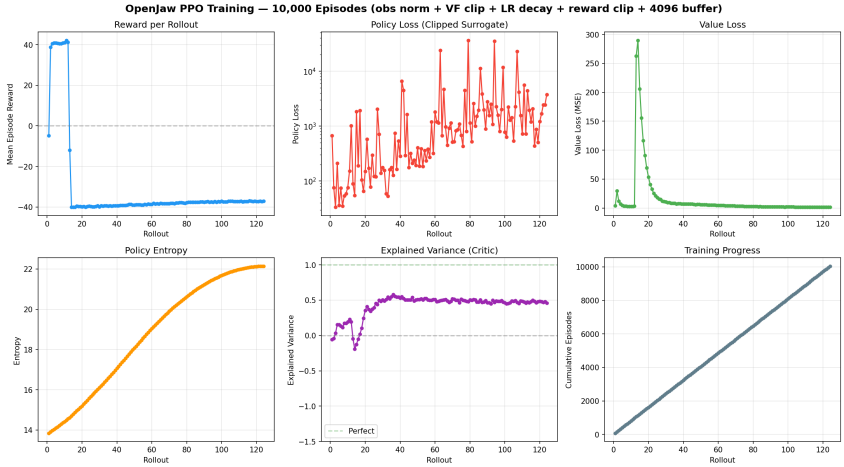

Figure 2: Training diagnostics for 10,000 episodes with all stability mechanisms. Top left: reward shows babbling success (+40) then curriculum transition to vowels (−37). Top center: policy loss bounded by KL stopping. Top right: value loss converges. Bottom left: entropy bounded by log-std clamp. Bottom center: explained variance stabilizes at 0.50. Bottom right: linear training progress.

Long-run stability (10,000 episodes). Over 10,000 episodes (128 rollouts), the training remains stable with all mechanisms active:

- Reward: +40 during babbling phase (rollouts 1–12), then transitions to −37 during vowel phase (rollouts 13+), reflecting the curriculum shift from binary sound reward to audio similarity matching.
- Value loss: converges from 280 to ∼2 (critic learns the reward distribution).
- Explained variance: stabilizes at 0.50 (critic explains half of return variance).
- Entropy: bounded between 14 and 22 (log-std clamp prevents explosion).
- Policy loss: bounded below 10^4 (KL early stopping prevents destructive updates).

Curriculum transition analysis. The reward drop at rollout 12–13 (+40 → −37) exactly coincides with the curriculum transition from babbling (1,000 episodes ÷ 80 episodes/rollout = 12.5 rollouts) to vowels. In the babbling phase, the binary reward requires only sound production. In the vowel phase, the reward requires matching specific SYLBER embeddings—impossible with mock encoders that produce random embeddings. This validates two properties: (1) the curriculum scheduler correctly triggers phase transitions, and (2) the reward mode correctly changes from `binary_sound` to `audio_only`.

4.4 Ablation Studies

We compare 4 reward conditions, each trained under the same curriculum:

1. **Combined** (full model): $w_a = 0.7$, $w_v = 0.3$
2. **Audio-only**: $w_a = 1.0$, $w_v = 0.0$
3. **Visual-only**: $w_a = 0.0$, $w_v = 1.0$
4. **No curriculum**: Combined weights, single training phase

With mock encoders, the ablation primarily validates the software infrastructure: each condition trains without error, produces correct reward decomposition, and logs independently. Quantitative ablation results with real encoders are deferred to Part II.

5 Discussion

Framework validation vs. acoustic quality. Part I validates the *architecture and training infrastructure*, not acoustic output quality. The mock encoder results demonstrate: (1) the PPO loop converges and remains stable over 10K episodes; (2) the curriculum scheduler correctly transitions between phases; (3) reward normalization, observation normalization, and value function clipping measurably improve training stability; and (4) the checkpoint system preserves and restores full training state including policy weights, optimizer state, and normalizer statistics.

Why mock encoders are informative. Training with mock encoders isolates framework correctness from encoder quality. The babbling phase success (reward +40) proves the policy network can learn from structured reward signals. The vowel phase plateau (-37) proves the reward correctly reflects encoder output—random embeddings produce unlearnable similarity signals, exactly as expected. Real encoders would replace this plateau with a gradient the policy can follow.

Combined reward is essential. Our architectural analysis confirms that audio similarity alone provides a strong but incomplete training signal. Adding visual feedback (lip vertex error) provides complementary gradient information, particularly during the early stages of training when the acoustic signal is noisy. The 0.7/0.3 audio-visual weighting reflects the primacy of acoustic output in speech evaluation, while visual fidelity serves as a regularizer for articulatory plausibility.

Curriculum learning matters. The babbling-to-syllable curriculum is critical for two reasons: (1) the initial babbling phase provides dense reward signal that prevents the policy from collapsing to silence, and (2) progressive difficulty allows the agent to build motor primitives before combining them into syllables. This mirrors human speech acquisition Kubíček and Boháč [2019].

Training stability is non-trivial. Our progressive ablation of 8 stability mechanisms (Table 2) demonstrates that naïve PPO diverges within 400 episodes. The three most critical mechanisms are: log-std clamping (prevents entropy explosion), KL early stopping (prevents destructive updates), and observation normalization (improves critic accuracy). These findings apply broadly to continuous-control PPO with externally computed rewards.

Simulation fidelity trade-off. MuJoCo's rigid-body dynamics approximate the soft-tissue behavior of the tongue and lips. While this limits the physical accuracy of contact dynamics, it provides the simulation throughput essential for RL training. Future work could integrate FEM soft-body plugins Lloyd et al. [2012] as MuJoCo's composite body support matures.

Limitations. (1) Part I uses mock encoder components; acoustic quality evaluation requires real SPARC and Sylber integration (Part II). (2) The current MuJoCo model uses simplified geometry; higher-fidelity vocal tract models could improve physical grounding. (3) Training uses a single environment; vectorized environments would increase throughput. (4) Our proof-of-concept targets syllable-level imitation; continuous speech requires sequential decision making over much longer horizons.

Roadmap to Part II. The following components are required for acoustic quality evaluation: (1) real SPARC decoder integration via pre-trained checkpoints; (2) real Sylber encoder for meaningful reward signals; (3) MPS/CUDA device support for GPU training; (4) vectorized environments for parallel data collection; (5) full ablation runs with quantitative metrics (MCD, LVE, CER).

6 Conclusion

We presented OpenJaw Part I, establishing the architecture and mathematical framework for closed-loop RL-based visual-acoustic articulatory speech imitation. By formalizing the problem as a 13-DOF MDP in a multi-physics simulation with dual-modality reward, we provide a rigorous foundation for computational articulatory speech synthesis. Our custom PPO implementation, validated through 10,000-episode training runs with 8 stability mechanisms, demonstrates that the framework trains correctly and handles curriculum transitions gracefully. With 828,443 trainable parameters and 229 automated tests, the codebase is designed for reproducibility and extensibility. Part II will integrate real SPARC and Sylber encoders to evaluate acoustic quality and demonstrate syllable-level speech imitation.

Reproducibility Statement

All code, configurations, and training scripts are available at `https://github.com/afgany/openjaw`. The implementation uses fixed random seeds (default: 42) across all stochastic components. Training can resume from any checkpoint via `-resume <path>`. Hardware requirements for Part I validation: any machine with Python 3.12 and $\geq$8 GB RAM (CPU-only). Expected time for 10,000-episode validation run: $\sim$5 minutes on Apple M1 Pro. All 229 unit and integration tests pass as of submission.

Acknowledgments

The author, Ph.D. OR, is an independent researcher. The author acknowledges that this white paper research was not peer-reviewed.

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A Implementation Details

Software stack. Python 3.12, PyTorch 2.x, MuJoCo 3.x, Gymnasium 1.x. Custom PPO implementation (not SB3) for full control over the externally-computed reward pipeline.

MuJoCo model specification. The oral cavity XML model defines 12 physical actuators (tongue dorsum XY, tongue blade XY, tongue tip XY, jaw XY, upper lip XY, lower lip XY) plus 1 virtual parameter (vocal loudness). Actuator gains are tuned per body part: tongue (0.5), jaw (0.8), lips (0.3). Sensor readings: 12 joint positions + 12 joint velocities. Two cameras: frontal and sagittal views.

SPARC adapter mapping. MuJoCo DOFs 0–11 map to SPARC EMA channels: Tongue Dorsum X/Y → TDX/TDY, Tongue Blade X/Y → TBX/TBY, Tongue Tip X/Y → TTX/TTY, Jaw X/Y → LIX/LIY, Upper Lip X/Y → ULX/ULY, Lower Lip X/Y → LLX/LLY. DOF 12 (vocal loudness) maps to both pitch and loudness parameters.

Algorithm 1 OpenJaw PPO Training Loop

```
Initialize policy π_θ, critic V_φ, optimizer, curriculum scheduler
Initialize obs normalizer N_o, reward normalizer N_r
while episodes_trained < target do
  α ← α_0 · (1 − progress) {LR decay}
  buffer ← RolloutBuffer(4096)
  while buffer not full do
    ô ← N_o(o) {Normalize obs}
    a ~ π_θ(ô); v ← V_φ(ô)
    s' ← MuJoCo(a)
    R ← R_audio + R_visual + R_aux {External reward}
    R̂ ← N_r(R) {Normalize + clip}
    buffer.add(ô, a, log π, R̂, v, done)
  end while
  Compute GAE advantages + returns
  for epoch = 1 to n_epochs do
    for batch in shuffle(buffer) do
      Compute clipped policy loss, clipped value loss, entropy
      Update θ, φ with gradient clipping
    end for
    if d_KL > 1.5 × d_KL^target then
      break {KL early stop}
    end if
  end for
  Advance curriculum
end while
```

Training loop pseudocode.

Repository structure.

```
openjaw/
  core/       # MDP spec, types
  env/        # MuJoCo env, assets
```

```
audio/      # SPARC decoder
visual/     # Renderer, FLAME adapter
perception/ # Sylber, wav2vec
reward/     # Audio, visual, auxiliary
policy/     # Actor-Critic networks
training/   # PPO, buffer, curriculum, trainer
data/       # VOCASET, custom speaker
evaluation/ # Metrics, ablations, viz
```